

Master of Disguise: Toward Reversible and Universal Facial Privacy Protection Service With Natural and Original Visual Semantics

Dezhi An, Jingshan Zheng^{1b}, Ruoyu Zhao^{1b}, Rushi Lan^{1b}, and Yushu Zhang^{1b}, *Senior Member, IEEE*

Abstract—Facial images, as sensitive biometric data, have raised increasing concerns about privacy leakage due to their widespread use in scenarios such as social networks, security surveillance, and smart devices. If a facial privacy protection method simultaneously possesses visual naturalness, perceptual originality, reversibility, and feature universality, it can effectively prevent identity leakage and maintain the visual usability of images for human recognition. In addition, recovery of the original image is supported when necessary. However, existing methods often succeed only in part and fail to achieve all four properties simultaneously. To address this limitation, we propose a facial privacy protection method that integrates image restoration with information hiding. The proposed method combines mosaic-based pixelation, super-resolution reconstruction, and information hiding to achieve reversible facial privacy protection while enabling high-fidelity recovery of the original image. We further implement a prototype to evaluate the feasibility and effectiveness of the proposed method. Experimental results demonstrate that it achieves strong performance in privacy protection, image quality, and high-fidelity recovery of the original image.

Index Terms—Image privacy protection, facial images, super resolution, image steganography.

Received 15 January 2026; revised 21 April 2026; accepted 12 June 2026. Date of publication 16 June 2026; date of current version 1 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62562005, in part by the Ganpo Talent Program of Jiangxi Province under Grant gpyc20240012, in part by the Outstanding Youth Fund Program of Jiangxi Province under Grant 20252BAC220008, in part by Jiangxi Key Research and Development Program under Grant 20261BCE310050, in part by the Science and Technology Research Project of the Jiangxi Provincial Department of Education under Grant GJJ2500409, in part by Lanzhou Science and Technology Plan Project under Grant 2023-1-53, in part by Anning Science and Technology Plan Project under Grant 2024-JB-5, and in part by the University-Level Innovative Research Team of Gansu University of Political Science and Law. (Corresponding author: Ruoyu Zhao.)

Dezhi An is with the School of Cyber Security and the Gansu Provincial Interdisciplinary Center for Digital Rule of Law, Gansu University of Political Science and Law, Lanzhou 730070, China (e-mail: adz6199@gsupl.edu.cn).

Jingshan Zheng is with the School of Cyber Security, Gansu University of Political Science and Law, Lanzhou 730070, China (e-mail: zhengjingshan@stu.gsupl.edu.cn).

Ruoyu Zhao and Yushu Zhang are with the Jiangxi Provincial Key Laboratory of Multimedia Intelligent Processing and the School of Computing and Artificial Intelligence, Jiangxi University of Finance and Economics, Nanchang 330013, China (e-mail: zhaoruoyu@jxufe.edu.cn; zhangyushu@jxufe.edu.cn).

Rushi Lan is with the Guangxi Key Laboratory of Trusted Software, Guilin University of Electronic Technology, Guilin 541004, China (e-mail: rslan2016@163.com).

The code is publicly available at <https://github.com/SSSun111/Reversible-Face-Privacy>.

Digital Object Identifier 10.1109/TDSC.2026.3704642

I. INTRODUCTION

WITH the rapid development of the digital age, images, as an important medium of information, are widely used in fields [1] such as social media, security surveillance, and smart devices. Facial images contain unique biometric features such as inter-pupil distance, cheekbone shape, and skin micro-textures. These characteristics have made facial images an irreplaceable form of biometric identification. These features have multi-dimensional, unstructured spatial distributions [2], significantly increasing privacy leakage risks. The incident of ClearView AI [3] involved unauthorized collection of facial data to build a recognition system. This case exposes the security risks of biometric feature information in uncontrolled scenarios. It is important to note that unauthorized use of facial feature data without encryption may pose privacy risks [4], including identity forgery and data misuse. The security vulnerabilities could cause irreversible harm to personal rights and trigger associated economic risks [5].

To this end, existing studies have proposed various protection methods to address potential threats such as unauthorized machine recognition. Early methods (e.g., mosaicing, blurring, and masking [6]) relied on the advantage of low computational complexity and achieved basic visual anonymization by disrupting the pixel structure of the facial region. However, these methods generate prominent boundaries that create visual perceptual cues, making privacy protection visually conspicuous [7]. This spatial consistency disruption causes perceptual conflicts. These conflicts violate the principle of imperceptibility [8] in privacy masking. Privacy-preserving transformations should not highlight protected regions or create visible artifacts. Such distortions reduce usability and may create extra privacy risks by indicating that sensitive information is being hidden.

Traditional encryption methods [9] achieve reversible mapping through pixel-level permutation and randomization. Unlike irreversible visual protection methods, they support key-based reversibility and offer algorithmic universality. However, noise-like encrypted images [10] completely break visual readability and semantic correlation [11], and fixed encryption strategies cannot support dynamic access control. In scenarios such as smart security, where data retrieval is required, these methods fail to meet the new demand for balancing privacy protection with data utility.

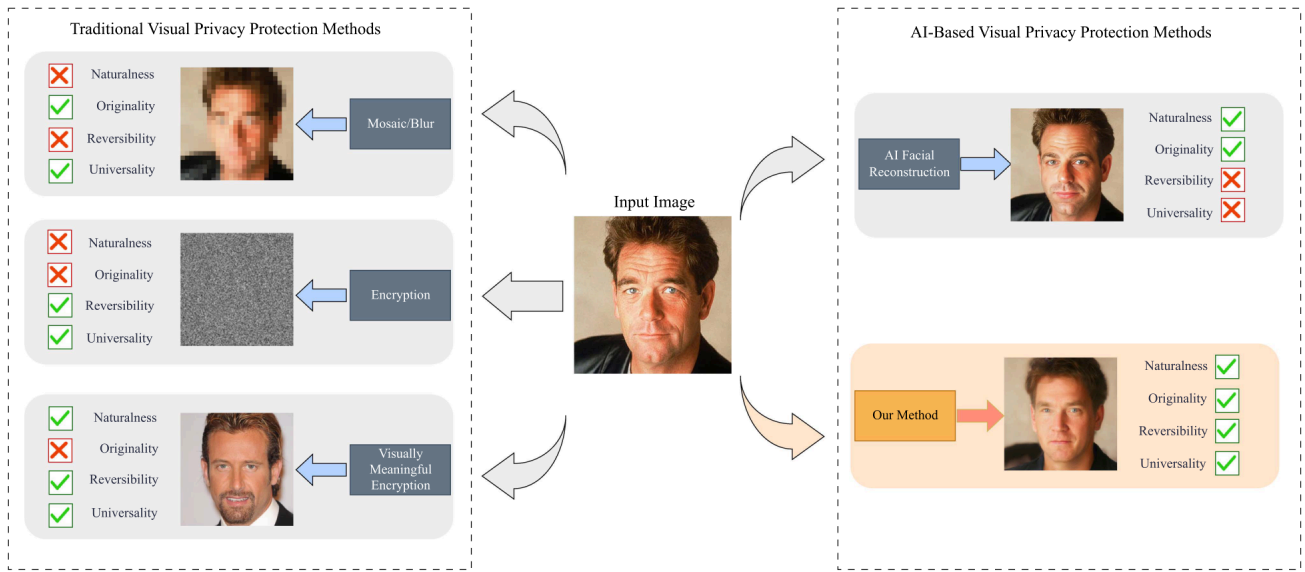


Fig. 1. Comparison between existing visual privacy-preserving methods and our method. Our method overcomes the shortcomings of existing methods in terms of naturalness, originality, reversibility, and universality of privacy protection.

Visual meaningful image encryption methods generate encrypted images by combining pre-encryption and discrete wavelet transform with a reference image [12]. These methods address the visual naturalness limitations of traditional encryption methods. The methods enhance concealment while maintaining visual plausibility. However, the appearance of the generated image is not directly related to the original image [13] but rather resembles the reference image [14]. This means that the encrypted image may visually differ significantly from the original, making it difficult to intuitively reflect the content and features of the original image. Therefore, it remains challenging to balance encrypted image naturalness with visual correlation to the original image.

Deep learning-based facial reconstruction methods can address the aforementioned issues. These methods employ hierarchical feature constraint networks to replace original facial features, achieving global visual consistency while balancing privacy strength and visual naturalness [15]. The current research bottlenecks mainly focus on the accuracy of cross-racial feature adaptation and the preservation of facial structure under unconstrained environments (e.g., facial pose diversity). There is an urgent need to develop cross-domain feature mapping and generalization optimization strategies to overcome the application limitations in real-world scenarios.

This study addresses the bottlenecks in current image privacy protection, such as visual naturalness, data originality, reversible recovery capability [16], [17], and scene applicability. It proposes an innovative privacy protection method that integrates image restoration and original information hiding. The method constructs a bidirectional reversible processing flow that efficiently masks sensitive information while maintaining visual naturalness and semantic integrity. Specifically, the image is first subjected to a mosaic-based pixelation operation to suppress sensitive details and is then reconstructed to the original resolution through super-resolution. The original image information

is subsequently embedded into the processed image, so that the resulting image remains visually correlated with the original while retaining the capability for high-fidelity recovery. As shown in Fig. 1, the proposed method addresses the limitations of prior methods by improving visual naturalness, supporting reversibility, and maintaining universality. This method thus provides a secure and efficient solution for privacy protection.

The main contributions of this paper are as follows:

- We analyze the inherent limitations of existing privacy protection methods. Based on this analysis, we propose an innovative method that integrates image restoration and information hiding technologies for the first time. This method successfully achieves the naturalness, originality, reversibility, and universality of facial privacy protection.
- We develop a technical prototype that encompasses two key components, namely super-resolution restoration and original information embedding. The prototype is designed to blur sensitive facial information, improve visual quality, and ensure secure recovery of the original data.
- We validate the proposed method using publicly available models. The results show that it achieves a favorable trade-off between privacy protection, image usability, and information recovery. The method demonstrates feasibility and provides a new solution for privacy protection tasks.

The remainder of this paper is organized as follows. Section II reviews related work on visual privacy protection. Section III presents the proposed method. Section IV describes the prototype implementation. Section V reports the experimental results and analysis. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Traditional Visual Privacy Protection Methods

Traditional visual privacy protection methods usually refer to directly processing original image or video data. The goal

is to reduce or hide sensitive information, thereby lowering privacy leakage risk. Privacy protection methods have been continuously improved, driven by growing demands in sensitive applications such as biometric recognition and identity authentication. Traditional visual privacy protection methods include image masking, traditional encryption, and visually meaningful encryption. These methods have addressed privacy threats in different scenarios and laid the foundation for subsequent research.

Early privacy protection methods disrupt the recognizability of facial biometric features through pixel-level operations [18]. Typical methods include mosaicing, blurring, and pixelation. These methods rely on local or global pixel perturbation strategies, achieving identity obfuscation by lowering image resolution or disrupting feature space correlations. Due to their low computational complexity, they were widely used in early surveillance systems. However, Wang and other researchers [19] pointed out that coarse-grained pixel modification irreversibly damages key semantic facial information. As a result, the processed image becomes unsuitable for input into intelligent analysis systems. With the advancement of deep fake methods, such as generative adversarial networks, simple spatial disturbances can be reverse-engineered to reconstruct the original facial features [20]. Nevertheless, such reconstruction should be viewed only as an imagined supplement rather than a reliable recovery.

Traditional encryption methods employ mathematical transformations to achieve data-level security protection. Symmetric encryption (e.g., AES [21]) and asymmetric encryption (e.g., RSA [22]) algorithms convert facial images into unreadable ciphertext. These methods provide information-theoretic security but compromise visual readability, limiting utility in post-processing and recognition tasks [23].

To address the limitations of the above methods, visually meaningful encryption has been proposed. This method encrypts the original image into a visually meaningful image rather than traditional texture or noise, achieving privacy protection while maintaining visual plausibility. Bao and Zhou [24] were the first to propose a new image encryption method. They altered the pixel positions and values through a pre-encryption process. The pre-encrypted image was then combined with a reference image using discrete wavelet transform. This process generated a visually meaningful encrypted image. The method enhanced both the security and concealment of the encrypted image [25]. This method addresses the visual shortcomings of traditional encryption methods. However, the trade-off between privacy protection strength and image quality remains a challenge in encryption methods.

In addition to the above methods, conventional reversible data hiding (RDH) and robust image-hiding methods also provide a relevant technical perspective for comparison. Recent studies have explored PVO-based reversible data hiding from different perspectives, including context-pixel-based and reference-pixel-based designs. Robust image-hiding methods have also been investigated in compressed or structured settings, such as BTC- and Hamming-code-based hiding and fuzzy-proximity-based embedding [26], [27]. However, these methods are mainly designed for general image hiding rather than facial privacy

protection with visually natural de-identification and authorized recovery.

Traditional visual privacy protection methods have laid an important foundation for privacy preservation. However, important limitations remain. Existing methods mainly emphasize confidentiality, concealment, or recoverability under specific settings. As a result, they do not provide a balanced solution that simultaneously supports privacy protection, visual usability, and recovery under authorized conditions.

B. AI-Based Visual Privacy Protection Methods

With the rapid advancement of artificial intelligence (AI) technologies, AI-based visual privacy protection methods have become an important research direction in image processing and privacy preservation. In facial image privacy protection, recent AI-based studies have mainly focused on generative face reconstruction and replacement. These methods typically use hierarchical feature constraints, latent-space manipulation, or diffusion-based generation to replace identity-sensitive facial features while preserving the overall visual style of the image.

Fang et al. [28] proposed a generative adversarial network (GAN)-based directional synthesis framework that enables the replacement of facial features while maintaining background consistency. This method incorporates multi-level feature fusion and constraint mechanisms to significantly alter identity features while preserving the visual naturalness of the image, thereby enhancing privacy protection.

Zhao et al. [29], [30] developed a GAN framework that performs privacy embedding in the latent space. The method leverages a feature pyramid network for hierarchical feature extraction and semantic-level adjustments, and it exploits the latent space modeling capability of StyleGAN. This design effectively suppresses identity information while preserving structural and visual consistency. This achieves a balance between image fidelity and privacy protection.

Shaheryar et al. [31] introduced IDiffuse, a diffusion model-based privacy protection method that unifies face anonymization and identity replacement. The method employs a dual-conditional diffusion model with identity mapping and multi-model feature extraction to guide the diffusion process. It enables controllable and high-quality identity replacement while preserving style and non-identity attributes.

In addition to these generative methods, recent studies have also explored adversarial identity obfuscation, reversible privacy protection, and invertible neural networks. Adversarial identity obfuscation mainly aims to mislead face recognition models through carefully designed perturbations. Reversible privacy protection and invertible neural networks place greater emphasis on the recoverability of protected content through reversible mappings or invertible transformations. Although these directions are relevant to privacy protection, each of them addresses only part of the problem. In facial privacy scenarios, they do not simultaneously provide visually natural de-identification for human observers and controllable recovery.

In summary, recent AI-based facial privacy protection methods have significantly improved privacy preservation while

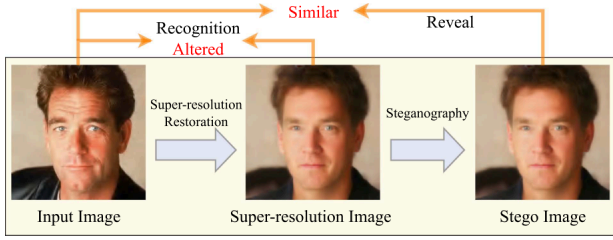


Fig. 2. The diagram of the privacy protection method. The word “similar” above the arrows means that the extracted secret image is close to the original one. It achieves high-fidelity reconstruction, although the process is not strictly lossless.

maintaining visual realism. Nevertheless, important limitations remain, especially for learning-based generative methods. Their effectiveness may still be sensitive to the distribution characteristics of the training data. When input images differ substantially from the training set in terms of ethnicity, pose, or other attributes, the privacy protection performance may degrade significantly. This limitation increases the risk of identity leakage and remains a major obstacle to practical deployment.

III. THE PROPOSED METHOD

The overall workflow of the proposed privacy protection method is illustrated in Fig. 2. The method consists of two core components, namely an image super-resolution restoration module and an image steganography module. These two modules work collaboratively to achieve naturalness, originality, reversibility, and universality in facial privacy protection. In the processing pipeline, the original facial image is first subjected to a mosaic-based pixelation operation for preliminary de-identification and feature weakening to reduce recognizability. Second, the image super-resolution restoration module generates a substitutive image with a style consistent with the original face. This process conceals the individual’s true identity while preserving visual naturalness and human perceptual usability. As a result, the protected image remains suitable for basic human perceptual tasks, such as counting the number of people or recognizing the approximate orientation of faces. Finally, the image steganography module employs a reversible information embedding mechanism to hide the original image within the generated image, enabling controllable recovery of identity information and high-fidelity reconstruction.

A. Image Super-Resolution Restoration

1) *Image Blur Reconstruction*: The image degradation step aims to achieve preliminary visual de-identification by suppressing structural information. This reduces the recognizability of sensitive facial images and prepares the image for subsequent reconstruction and information embedding. In this stage, mosaic processing is introduced, where the original pixel regions are replaced with block-wise average colors. This operation weakens local facial textures and geometric structures. It is important to clarify that this degradation operation is not designed to function as an independent anonymization method. Instead, it is

used solely as an intermediate step within the proposed method to attenuate identifiable facial features before the subsequent reconstruction and embedding stages. The mosaic operation can be expressed as:

$$I_{LR} = M_b(I), \quad (1)$$

where $M_b(\cdot)$ denotes the mosaic operation with block size b . The parameter b controls the degree of blurring: a larger b results in fewer image details and stronger privacy protection. Given an input image $I \in \mathbb{R}^{H \times W \times C}$, the image is divided into several blocks of size $b \times b$, where b is the adjustable block size parameter. For each image block $B_{i,j} \subset I$, the following operation is performed:

$$\bar{B}_{i,j} = \frac{1}{|B_{i,j}|} \sum_{(x,y) \in B_{i,j}} I(x,y). \quad (2)$$

The average color of the current block is calculated and used to replace all pixels within the block, resulting in a degraded mosaic image.

The super-resolution image reconstruction aims to restore the blurred input image I_{LR} into a high-resolution image I_{SR} with natural visual appearance and consistent:

$$I_{SR} = \mathcal{R}(I_{LR}; \theta), \quad (3)$$

where $\mathcal{R}(\cdot)$ denotes a learnable reconstruction mapping function controlled by the parameter θ .

2) *Loss Function*: During the super-resolution process, multiple loss functions are introduced to jointly optimize the reconstruction quality at both pixel and semantic levels. Each loss function constrains the quality of the generated image from a different perspective.

The downsampling consistency loss is designed to ensure that the generated image remains consistent with the input image after downsampling. It is defined as:

$$\mathcal{L}_{\text{down}} = \|DS(I_{SR}) - I_{LR}\|_p^p, \quad (4)$$

where $DS(\cdot)$ denotes the downsampling operation, which typically uses bicubic interpolation or Gaussian blur downsampling. This loss simulates a reverse constraint of the image degradation process and is suitable for weakly supervised training scenarios where paired data are limited.

Traditional pixel-level losses struggle to accurately capture high-frequency details and semantic structures of images. To address this, a perceptual loss based on deep feature space is introduced, defined as:

$$\mathcal{L}_{\text{percep}} = \sum_l \|\varphi_l(I_{SR}) - \varphi_l(I_{GT})\|_2^2, \quad (5)$$

where $\varphi_l(\cdot)$ represents the intermediate features extracted from the l -th layer of a pre-trained network, and I_{GT} denotes the original high-resolution image. This loss emphasizes semantic consistency and can significantly improve the perceptual quality of the images.

When using generative models, a prior regularization term is introduced to stabilize the image generation process and

constrain the latent variable distribution, defined as:

$$\mathcal{L}_{\text{latent}} = \left\| \|z\|_2 - \sqrt{d} \right\|^2, \quad (6)$$

where $z \in R^d$ denotes a latent space vector, and its norm is constrained to be close to the radius $\sqrt{d}$ of a Gaussian hypersphere. This term can be regarded as a constraint on the sampling range in the latent space, preventing the generator from producing images that lie outside the manifold of real images.

The final optimization objective is formulated by combining multiple loss functions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{down}} + \lambda_1 \mathcal{L}_{\text{percep}} + \lambda_2 \mathcal{L}_{\text{latent}}, \quad (7)$$

where λ_1 and λ_2 are tunable hyperparameters used to balance the weights of different loss terms. In practical training, different weight combinations can be set according to the model complexity and task objectives to achieve a trade-off between reconstruction quality and convergence speed.

B. Image Steganography

1) *Image Embedding Extraction*: The image steganography module serves as a key component of the overall method to ensure information security. It is responsible for effectively hiding and accurately extracting embedded information while preserving the structural integrity and visual naturalness of the image [32]. By considering the imperceptibility of embedding [33], the integrity of image quality [34], and the scalability of the overall structure, this module meets the privacy protection requirements in complex scenarios.

Let the externally visible carrier image be I_{SR} , and the target image to be embedded be $I \in \mathbb{R}^{H \times W \times C}$. The steganographic embedding function is denoted as $S(\cdot)$, and the decoding recovery function as $D(\cdot)$. The overall input-output process of the module can be defined as follows:

The embedding operation on the carrier image is defined as:

$$I_E = S(I_{SR}, I), \quad (8)$$

where I_E denotes the stego image after embedding, which should preserve a high degree of structural and visual consistency with the original carrier image I_{SR} .

The hidden image is extracted from the stego image:

$$\hat{I} = D(I_E), \quad (9)$$

where $\hat{I}$ denotes the recovered hidden image extracted from the stego image. It should approximate the original image I as closely as possible in terms of pixel structure and content information.

2) *Loss Function*: To ensure high-quality embedding and accurate recovery in the image-in-image task, it is necessary to jointly optimize key metrics in both the embedding and extraction processes. The loss function is designed to guide the model in achieving the following three objectives: (1) preserving the visual consistency of the embedded image to make it imperceptible, (2) retaining as much structural information of the target image as possible to ensure accurate reconstruction, and (3) balancing multi-scale perceptual quality and structural fidelity during optimization.

Based on the above objectives, three types of loss functions are constructed:

The embedded image should be as visually close as possible to the original carrier image to enhance imperceptibility and reduce the risk of exposure to external detection mechanisms. To achieve this, a fidelity loss is constructed by combining pixel-level error with a structural similarity index, defined as:

$$\mathcal{L}_{\text{rec}} = \|I_E - I_{SR}\|_1 + \lambda_1 (1 - \text{SSIM}(I_E, I_{SR})), \quad (10)$$

where I_E denotes the stego image, I_{SR} is the original carrier image, λ_1 is a balancing coefficient, and SSIM measures structural consistency.

To ensure the accuracy of the extracted image, the difference between the recovered image and the original target image is measured. The L1 norm is typically used as the reconstruction loss, defined as:

$$\mathcal{L}_{\text{hid}} = \|\hat{I} - I\|_1, \quad (11)$$

where $\hat{I}$ denotes the recovered image, and I represents the original target image before embedding.

To further improve the detail quality and semantic consistency of image restoration, this module introduces a perceptual loss based on feature space. The method is especially beneficial for high-resolution images or structurally complex scenarios. By constraining the differences in feature maps extracted from intermediate layers of a pre-trained image classification network, the model is enhanced in preserving textures, edges, and semantic structures. The loss is defined as:

$$\mathcal{L}_{\text{perc}} = \sum_l \left\| \varphi_l(\hat{I}) - \varphi_l(I) \right\|_2^2, \quad (12)$$

where $\varphi_l(\cdot)$ represents the feature maps extracted from the l -th layer of the network. This loss effectively compensates for the limitations of pixel-level errors in capturing perceptual information. During training, appropriate regularization terms (e.g., weight decay) can also be introduced to improve the model's generalization ability and suppress overfitting.

The total loss function is composed of the above three terms, and the overall optimization objective is defined as:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{hid}} + \gamma \mathcal{L}_{\text{perc}}, \quad (13)$$

where α , β , and γ are weighting coefficients for each loss term. These coefficients should be set according to specific experimental requirements and data distribution conditions.

All loss-balancing hyperparameters (e.g., λ_1 , λ_2 , α , β , and other similar weighting coefficients) adopt conventional values commonly used in prior literature. These parameters serve strictly to stabilize training by balancing different loss components. They do not affect the task-level behavior, privacy strength, or reversibility performance of the method. Consequently, they are kept fixed across the experiments.

IV. PROTOTYPE CONSTRUCTION

A technical prototype is designed and implemented to verify the feasibility and effectiveness of the proposed image privacy

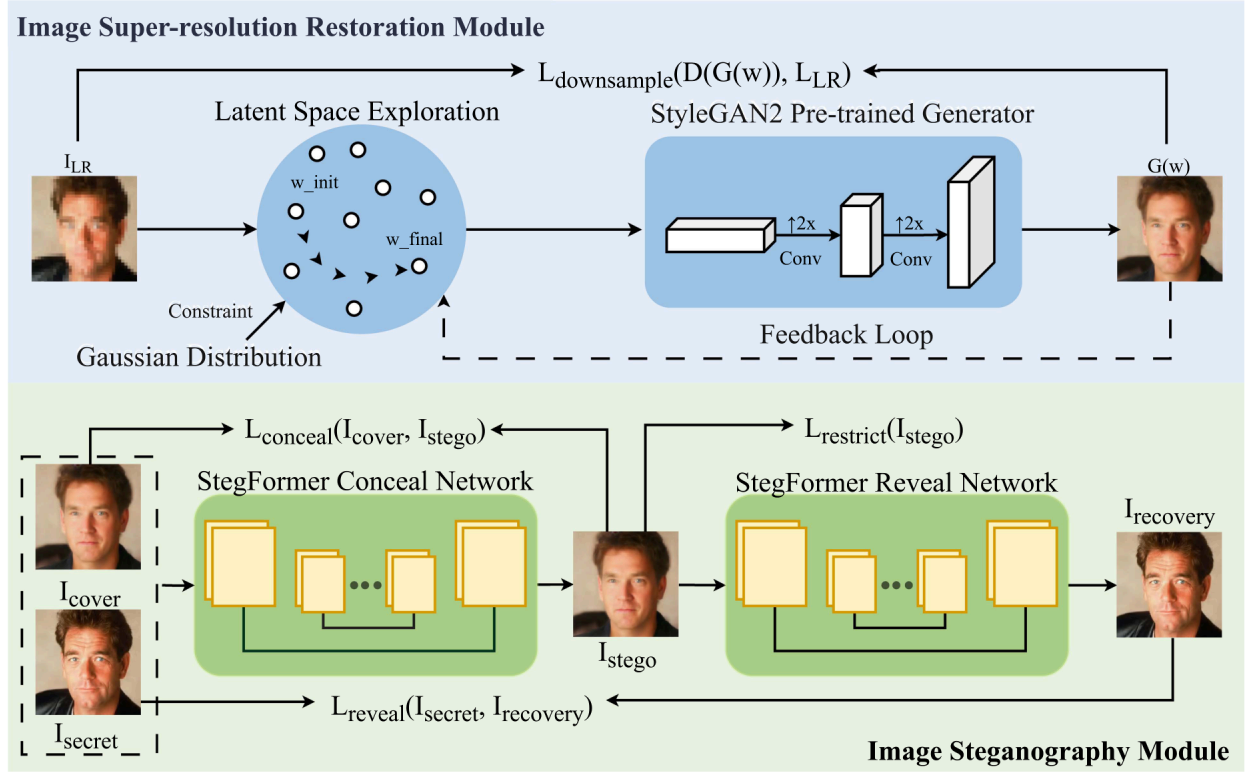


Fig. 3. Detailed architecture of the proposed privacy protection method.

protection method. The prototype is intended to remove identity information from sensitive images and perform efficient steganography, thereby ensuring image privacy while enhancing the robustness of information hiding. As illustrated in Fig. 3, the overall architecture of the prototype consists of two core modules: an image super-resolution restoration module and an image steganography module. To demonstrate the applicability of the method, the PULSE [35] model is adopted to implement the super-resolution module, and the StegFormer [36] model is employed for the steganography module. In the current prototype, reversibility is achieved by decoding the original information embedded in the protected image, rather than by exact inversion of the super-resolution module. Thus, the proposed method demonstrates reversibility through high-fidelity recovery, instead of relying on a strict mathematical guarantee of exact lossless inversion for every module. It should be noted that these two models are used only as illustrative implementations. The method does not rely on the specific architectures or characteristics of these models. Other models may also be considered for these roles if they provide the capabilities of super-resolution reconstruction and information hiding.

A. Image Super-Resolution Restoration Module

This module aims to perform high-quality reconstruction of processed low-resolution sensitive images to enhance visual fidelity and image detail levels. It provides high-quality and detail-rich inputs for subsequent image steganography. In our prototype, this module adopts the PULSE model [35] for

super-resolution reconstruction. Unlike traditional supervised methods that rely on paired low- and high-resolution images for training, PULSE performs reconstruction through latent space optimization based on a pretrained generative model. In the proposed method, the mosaiced image serves as the degraded input, and the reconstructed high-resolution image is then passed to the steganography module for subsequent information embedding.

In the specific implementation, assume that a pre-trained generative model $G(\cdot)$ can map a latent vector w to a high-resolution image $G(w)$. To ensure that the generated image is consistent with the input low-resolution image I_{LR} while preserving naturalness, the following optimization objective is defined:

$$\min_w L_{\text{downsample}}(D(G(w)), I_{LR}) + \lambda \mathcal{R}(w), \quad (14)$$

where $D(\cdot)$ denotes the downsampling operator (e.g., bicubic interpolation). $\mathcal{R}(w)$ is a regularization term for the latent vector. It constrains the distribution to prevent the generated image from deviating from the natural image space, thereby ensuring perceptual realism. $L_{\text{downsample}}$ is the loss term between the downsampled generated image and the input low-resolution image. It is typically expressed in the mean squared error (MSE) form as:

$$\mathcal{L}_{\text{downsample}} = \|D(G(w)) - I_{LR}\|_2^2. \quad (15)$$

The original image is first processed with a mosaic operation to produce a low-resolution image, which is then fed into the model. To meet the requirements of the generative model, the

Algorithm 1: Super-Resolution Reconstruction Module.

-
- Require:** Low-resolution input image I_{LR}
Ensure: High-resolution output image $G(w)$
1: Input: Provide the original low-resolution sensitive image I_{LR} ;
2: Preprocessing: Perform grayscale normalization and resize adjustment on the input image;
3: Latent Vector Initialization: Initialize the latent vector w randomly or based on mean values;
4: Image Generation: Use the generator model $G(w)$ to generate the high-resolution image;
5: Downsampling Alignment: Downsample the generated image and compare it with the input low-resolution image;
6: Loss Computation and Optimization: Calculate the loss function and update the latent vector;
7: Output: Obtain the final high-resolution image $G(w)$.
-

input image undergoes grayscale normalization and resizing before entering the latent space optimization stage, ensuring consistency with the input distribution of the pre-trained model. Second, a latent space search is performed to iteratively optimize the latent vector w , generating a high-resolution image $G(w)$ that satisfies the above optimization constraints. This module outputs a high-resolution image that remains consistent with the input at the downsampling level and serves as input for the subsequent image steganography module, enabling a closed-loop workflow.

During optimization, the latent vector w is updated using spherical projected gradient descent, constraining the latent vector search to the surface of a high-dimensional Gaussian hypersphere. This ensures that the generated image remains on the natural image manifold, avoiding artifacts or unrealistic structures. In addition, the multi-scale noise inputs introduced by StyleGAN during the generation process further enhance detail expression. As a result, the optimized high-resolution image not only maintains high consistency with the input low-resolution image after downsampling but also exhibits rich textures and perceptual naturalness. The detailed procedure is presented in Algorithm 1.

B. Image Steganography Module

This module is designed to embed sensitive secret images into carrier images, achieving efficient hiding and high-fidelity extraction of secret information while preserving the visual quality of the carrier image [37], [38], [39]. In the technical prototype, the StegFormer model is adopted as the implementation of the steganographic method. By utilizing an end-to-end encoder-decoder architecture, the module performs both embedding and recovery of secret images. In the current prototype, the reverse recovery path of the proposed method is realized through this embedding-extraction mechanism, rather than through direct inversion of the visual reconstruction module.

The StegFormer model is built upon a U-Net backbone and consists of two components: a Conceal Network and a Reveal

Network, as illustrated in Fig. 4. The Conceal Network performs the embedding of secret images, while the Reveal Network handles the decoding and reconstruction of the hidden information. The overall design of the model considers both local and global feature modeling, supporting high-capacity information hiding and high-fidelity image recovery.

In the implementation, the input carrier and secret images are respectively denoted as I_{cover} and I_{secret} . The goal of the model is to generate the stego image I_{stego} and to recover the secret image $I_{recovery}$ during the extraction phase. To ensure the quality and integrity of information hiding, StegFormer integrates the following loss function during training:

$$\mathcal{L}_{total} = \mathcal{L}_{conceal} + \alpha \mathcal{L}_{reveal} + \beta \mathcal{L}_{restrict}, \quad (16)$$

where α and β are weighting coefficients.

$\mathcal{L}_{conceal}$ represents the difference loss between the stego image and the carrier image. To enhance robustness, it is typically formulated using the Charbonnier Loss, defined as:

$$\mathcal{L}_{conceal} = \sqrt{\|I_{stego} - I_{cover}\|^2 + \epsilon^2}. \quad (17)$$

$\mathcal{L}_{reveal}$ represents the difference loss between the recovered image and the secret image. It is also defined using the Charbonnier loss, as follows:

$$\mathcal{L}_{reveal} = \sqrt{\|I_{recovery} - I_{secret}\|^2 + \epsilon^2}. \quad (18)$$

$\mathcal{L}_{restrict}$ is a constraint loss used to enforce the pixel range of the stego image, preventing pixel overflow during training and ensuring the generalization performance of the model in real-world scenarios. It is defined as:

$$\mathcal{L}_{restrict} = \max(0, I_{stego} - 1)^2 + \max(0, -I_{stego})^2, \quad (19)$$

where ϵ is a smoothing factor introduced to avoid numerical instability.

StegFormer introduces the Channel Adaptive Transformer Block (CATB). This block enables dynamic adjustment and fusion of inter-channel information to optimize information flow during hiding and recovery. The CATB module combines local window attention with a channel adaptation strategy, enhancing the model's ability to capture fine-grained features and contextual relationships. In addition, the bottleneck layer incorporates the Global Enhancement Block (GEB). This block further expands the model's receptive field to capture long-range dependencies, thereby improving the integrity and robustness of the recovered images. To enhance the model's applicability in real-world environments, a normalizing training strategy is adopted. This strategy integrates the constraint loss with gradient clipping to mitigate distortion issues caused by pixel values exceeding the valid range. It also ensures the stability of stego images under complex channel transmission conditions.

During training, the input carrier image and secret image are first grayscale-normalized and resized to match the input requirements of the model. They are then concatenated along the channel dimension and fed into the Conceal Network, which outputs the stego image. The stego image is passed through a constraint operation to ensure that pixel values remain within the visible range. It is then used as the input to the Reveal Network

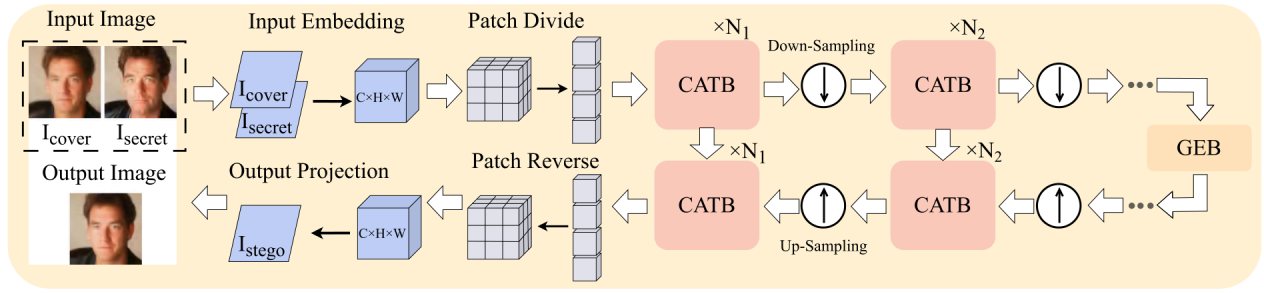


Fig. 4. Full components of the StegFormer.

Algorithm 2: Steganography Embedding and Recovery Module.

- Require:** Original cover image I_{cover} , original secret image I_{secret}
- Ensure:** Stego image I_{stego} and recovered secret image $I_{recovery}$
- 1: **Cover Image Input:** Input the original cover image I_{cover} ;
 - 2: **Secret Image Input:** Input the original secret image I_{secret} ;
 - 3: **Preprocessing:** Perform grayscale normalization and resize adjustment on the input images;
 - 4: **Input Concatenation:** Concatenate I_{cover} and I_{secret} along the channel dimension;
 - 5: **Stego Image Generation:** Generate the stego image I_{stego} using the Conceal Network;
 - 6: **Value Clamping:** Apply pixel value constraints to ensure the stego image is within a valid image representation range;
 - 7: **Secret Image Recovery:** Input the stego image into the Reveal Network to reconstruct the secret image $I_{recovery}$;
 - 8: **Loss Calculation and Optimization:** Compute the losses $\mathcal{L}_{cover}$, $\mathcal{L}_{secret}$, and $\mathcal{L}_{restrict}$ and update the network parameters accordingly;
 - 9: **Output:** Output the final stego image and the reconstructed secret image.
-

for decoding and reconstruction of the secret image. The detailed procedure is presented in Algorithm 2.

V. EXPERIMENT

A. Experimental Setup

We adopt two high-quality face image datasets, CelebA-HQ and FFHQ, which contain 30,000 and 70,000 high-resolution face images at 1024×1024 resolution, respectively. From these datasets, we select 250 images to evaluate the method's performance in terms of naturalness, originality, and reversibility. In addition, we use a separate set of 30 images with extreme expressions, large poses, partial occlusions, or individuals from minority ethnic groups to assess the method's universality.

In the experiments, images are first anonymized using a mosaic method with adjustable block sizes ($b = 10, 20, 30$) to control de-identification strength by replacing each block's pixels with their average color. The processed images are then input to the proposed method for super-resolution reconstruction and information embedding, enabling high-fidelity recovery while preserving privacy. In the current prototype, all protected images are stored and processed in a lossless format before decoding. Accordingly, the recovery experiments assume that the protected images remain intact prior to authorized extraction, rather than being subjected to post-processing distortions such as lossy recompression or platform-level modification.

Reproducibility is enhanced by providing detailed training configurations and implementation settings of the proposed method. The PULSE model is directly adopted as a pre-trained super-resolution module to reconstruct high-resolution facial details from the mosaic-degraded inputs, without any additional fine-tuning. The StegFormer-based steganography module is trained for 6000 epochs using the AdamW optimizer. The initial learning rate is set to $1e-3$ and gradually decayed to $1e-6$ following a cosine annealing schedule to ensure stable convergence. All experiments are conducted on a vGPU server equipped with 32 GB of GPU memory, and the entire prototype is implemented in PyTorch.

B. Naturalness Evaluation

The visual naturalness of the generated images is comprehensively evaluated using multiple metrics. These include entropy-based statistical measures and perceptual quality metrics such as FID and LPIPS. Entropy-based evaluation considers both the global entropy difference and the proportion of local entropy anomaly regions. Global entropy reflects the overall information richness of an image. The local entropy anomaly proportion indicates potential structural degradation, such as residual mosaic artifacts or excessive smoothing. A combined naturalness score is then obtained to quantify the perceptual realism of the reconstructed outputs. Perceptual quality is further assessed using FID and LPIPS. FID evaluates fidelity from the perspective of deep feature distributions, whereas LPIPS measures perceptual similarity between images.

The experiments are conducted under different anonymization strengths corresponding to mosaic block sizes of 10, 20, and 30. Fig. 5 shows the distribution of naturalness scores. Most

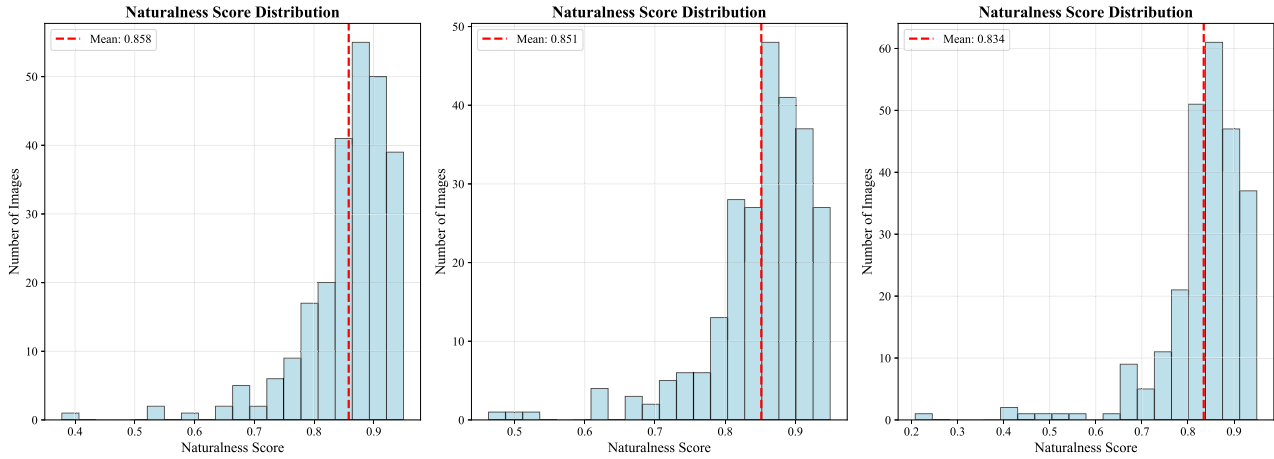


Fig. 5. Distribution map of naturalness scores(block_size=10, 20, 30).

TABLE I
PERCEPTUAL EVALUATION OF IMAGE QUALITY

Method	FID	LPIPS
Our method (block_size=10)	47.82	0.4858 $\pm$ 0.0599
Our method (block_size=20)	49.81	0.4921 $\pm$ 0.0631
Our method (block_size=30)	52.18	0.4976 $\pm$ 0.0565

TABLE II
STRUCTURAL SIMILARITY AND FACE REGION SIMILARITY RESULTS

Method	Structural similarity	Face region similarity
Our method (block_size=10)	0.9463	0.9126
Our method (block_size=20)	0.9413	0.9086
Our method (block_size=30)	0.9334	0.8987

reconstructed images achieve scores above 0.8, indicating that the proposed method consistently preserves high visual quality across various degradation conditions. The FID and LPIPS results are reported in Table I. The proposed method achieves consistently low FID values and stable LPIPS performance. These results indicate that reconstruction quality remains robust under varying degradation levels.

C. Recognizability and Structural Analysis

A multi-level originality analysis metric is introduced to evaluate the structural fidelity and identity recognizability of the protected images. Under low-resolution conditions, structural similarity indices compute the consistency of local morphology and texture structures. A facial key-region similarity metric measures the recognition accuracy of critical facial areas. When certain images fail to detect facial regions accurately, the method defaults to a full-image similarity evaluation to maintain metric consistency. Fig. 6 shows the distributions of the structural similarity and face region similarity metrics under different block size settings, and Table II summarizes the corresponding

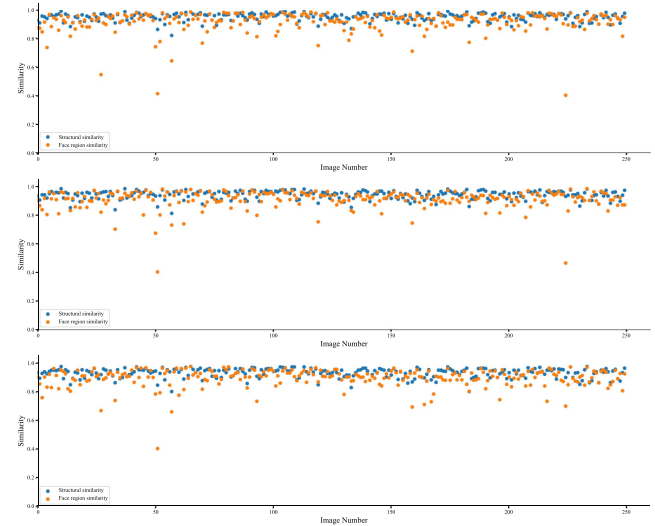


Fig. 6. Structural similarity and face region similarity distribution (block_size=10, 20, 30).

statistical results. These results demonstrate that the proposed method consistently achieves high scores on both metrics, confirming its applicability and effectiveness in image protection tasks.

A small number of outlier points are observed, mainly due to challenging samples such as extreme occlusions or pose variations. These cases are rare, and regenerating the results often improves similarity performance.

D. Reversibility Evaluation

Quantitative metrics are introduced to systematically evaluate the reconstruction capability of the proposed method during the information recovery stage. These metrics cover both luminance fidelity and structural restoration. The peak Signal-to-Noise Ratio in the Y channel (PSNR_Y) and the structural similarity index (SSIM) are used to quantify the luminance information and structural fidelity of the recovered images. In addition,

TABLE III
QUANTITATIVE EVALUATION OF COVER-STEGO AND RECOVERED-SECRET IMAGE PAIRS

Method	PSNR_Y		SSIM		RMSE		MAE	
	Cover	Secret	Cover	Secret	Cover	Secret	Cover	Secret
Our method (block_size=10)	50.96	55.11	0.986	0.998	1.088	1.087	0.768	0.846
Our method (block_size=20)	51.01	55.12	0.986	0.998	1.083	1.086	0.765	0.845
Our method (block_size=30)	51.08	55.15	0.986	0.998	1.076	1.084	0.761	0.844

we employ Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) to comprehensively measure the overall reconstruction error. Together, these four metrics characterize the restoration performance between recovered and original secret images, allowing an empirical evaluation of the reversibility of the method. In this work, high-fidelity recovery means that the recovered image remains highly consistent with the original one in both visual appearance and quantitative metrics, rather than being strictly lossless or bit-exact.

Table III presents the experimental results. Setiadi et al. [40] studied image steganography quality assessment. They found that images with PSNR values above 40 dB and SSIM values close to 1 usually show imperceptible visual changes, indicating good imperceptibility and reconstruction capability. In our experiments, the recovered images achieve PSNR_Y values above 50 dB and SSIM values above 0.986 for all block size settings. These values exceed the evaluation criteria by a large margin, indicating that the proposed method supports high-fidelity recovery during information embedding and extraction. This reversibility is demonstrated empirically, rather than guaranteed by a strict mathematical inversion of the entire method.

To further verify the embedding feasibility of the proposed method, the complete embedding and recovery pipeline is applied to all 280 test images used in the experiments. In addition, it is applied to all 30,000 images in the CelebA-HQ dataset. No embedding or decoding failures are observed across any of the evaluated samples. Higher perceptual similarity between the original and generated images may provide additional redundancy for information embedding. However, the recovery process in the proposed method does not depend on such similarity. The embedding and extraction procedures remain stable even in low-similarity cases, as further demonstrated by the large-scale feasibility verification described above.

E. Effectiveness Analysis of Identity Camouflage

A face verification experiment is performed to evaluate the identity camouflage effectiveness of the generated images. Specifically, facial features are extracted from each original image and its corresponding protected image using the face_recognition library. The Euclidean distance between the two feature vectors is then computed to determine whether the two images are recognized as belonging to the same identity. A threshold of 0.5 is used for identity verification.

Fig. 7 presents the results. Under the threshold of 0.5, all Euclidean distances between the protected and original images are higher than the discrimination standard, and no image pair is

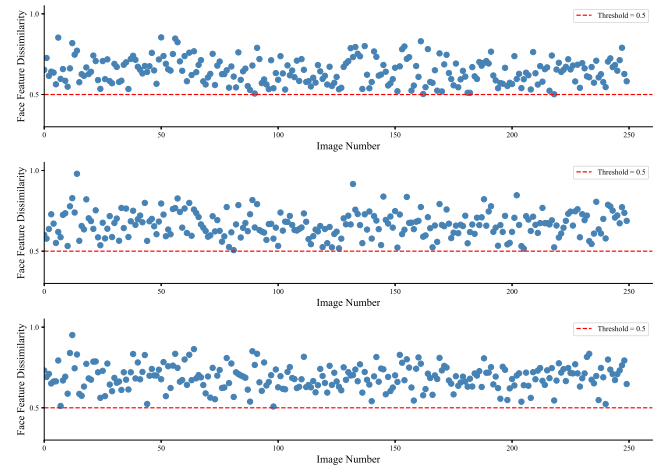


Fig. 7. Face feature dissimilarity distribution statistics (block_size=10, 20, 30).

identified as the same identity. These findings confirm that the proposed method effectively reduces facial recognizability after protection and achieves reliable identity camouflage.

F. Universality Analysis

A test set containing diverse face samples is constructed to verify the adaptability in complex scenarios. The set includes faces with extreme expressions, large pose variations, partial occlusions, and samples from different ethnic groups to simulate non-ideal conditions in real-world applications. Performance degradation in high-complexity scenarios is evaluated by comparing the proposed method with various existing methods to assess robustness and generalization capability.

As shown in Fig. 8, some existing methods often produce noticeable artifacts, color distortions, or structural integrity issues when processing samples with extreme facial expressions or abnormal skin tone rendering. In contrast, the proposed method maintains high reconstruction quality under various complex conditions, prevents significant distortions, and reduces the risk of identity information leakage.

To further validate the universality of the proposed method, additional experiments are conducted on the Realistic-Face-Portrait-1024px dataset. This dataset contains high-resolution facial images with diverse appearances and natural visual characteristics. Unlike more controlled datasets, this dataset better reflects real-world image distributions and introduces additional

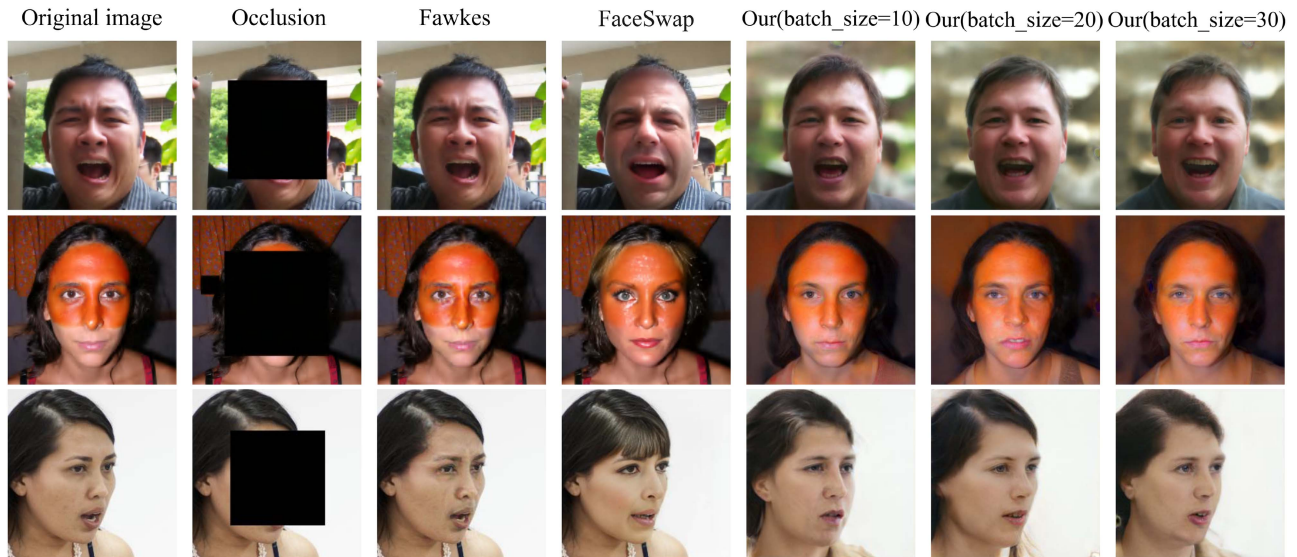


Fig. 8. Comparison of face privacy protection methods.

TABLE IV
QUANTITATIVE RESULTS OF THE PROPOSED METHOD WITH DIFFERENT BLOCK SIZES

Method	Naturalness score	Structural similarity	Face region similarity
Our method (block_size=10)	0.8591	0.9464	0.9289
Our method (block_size=20)	0.8533	0.9426	0.9256
Our method (block_size=30)	0.8388	0.9351	0.9246

challenges for facial structural preservation and visual naturalness. As reported in Table IV, the proposed method achieves consistently high naturalness, structural similarity, and face-region similarity under different block size configurations. Slight variations are observed as the block size increases. However, the overall performance degradation remains limited, indicating stable behavior across different parameter settings and data distributions.

G. Security Analysis

To evaluate the security of the proposed method, this part analyzes its resistance to unauthorized extraction under different threat models. It is assumed that an attacker can access the stego-image and is aware of the existence of hidden information. In the current prototype, reliable recovery is achieved under controlled conditions where the protected image remains intact prior to decoding. Distortions introduced during storage, transmission, or platform-level recompression may corrupt the embedded information and consequently degrade or even prevent recovery. This behavior is common in image-hiding methods and is consistent with the design objective of the proposed method, in which unauthorized modification of the protected image should invalidate the recovery process. Under this setting,

TABLE V
EXTRACTION PERFORMANCE UNDER PARAMETER-PERTURBED DECODERS

Parameter perturbation level	PSNR	SSIM	RMSE	MAE
Original	55.11	0.998	1.087	0.846
0.1%	40.62	0.964	3.311	2.693
0.3%	33.93	0.809	6.446	5.294
0.5%	30.28	0.542	11.198	8.795
1.0%	25.57	0.293	21.075	16.631
2.0%	14.92	0.056	65.158	49.929

security is assessed from two perspectives, namely resistance to unauthorized or approximate decoders and robustness under incorrect private-key conditions.

1) *Security Against Unauthorized Decoders*: A strong attack scenario is considered to evaluate the security of the proposed method against unauthorized decoders. In this scenario, the attacker is aware of the extraction model architecture and attempts to recover the hidden image using a nearly identical decoder. Specifically, the same StegFormer architecture is employed, while slight perturbations are introduced to the model parameters to simulate a parameter-approximation attack.

In this experiment, the stego-image is fed into decoders with progressively perturbed parameters, and the corresponding extracted images are evaluated using PSNR, SSIM, RMSE, and MAE. The experimental results are summarized in Table V. As the parameter perturbation increases, the extraction quality degrades rapidly. Even under minor perturbations, the extracted results become severely distorted and exhibit noise-like characteristics, with PSNR and SSIM decreasing significantly. These results indicate that the extraction process is highly sensitive to the decoder parameters. Unauthorized decoders, even when sharing the same architecture but with mismatched parameters, fail to recover meaningful visual or structural information from the stego-image. This indicates that meaningful extraction

TABLE VI
EXTRACTION PERFORMANCE UNDER DIFFERENT PRIVATE KEY CONDITIONS

Key condition	PSNR	SSIM
Correct key	33.79	0.830
Similar key	6.88	0.008
Wrong key	6.87	0.008

without authorization is practically infeasible under decoder mismatch conditions.

2) *Key-Based Security Analysis*: This part evaluates the security of the proposed method under a key-based protection mechanism. In the main prototype, the embedded information is not encrypted before hiding. Instead, security mainly relies on the learned embedding mechanism and the requirement for correct decoder parameters during recovery. To further strengthen resistance against unauthorized extraction, a private-key-based protection layer is introduced as an optional security enhancement. It is emphasized that this mechanism is not a core component of the method. The private key is composed of two components, namely a shuffling key and a multiplication key. The shuffling key permutes image blocks, whereas the multiplication key applies element-wise multiplication with randomly generated ± 1 values. To assess the effectiveness of this protection, extraction experiments are conducted using incorrect private keys while all other components remain unchanged.

As reported in Table VI, extraction with similar or incorrect private keys results in PSNR values around 7 dB and SSIM values close to zero, indicating that no meaningful visual or structural information can be recovered. These results demonstrate that incorrect or partially incorrect private keys effectively prevent meaningful recovery and provide additional protection against unauthorized extraction.

H. Comparative Analysis With Existing Methods

1) *Quantitative Evaluation*: To comprehensively evaluate the performance of the proposed method against existing privacy-preserving methods, both objective metrics and subjective perceptual assessments are employed in the comparative experiments. Since direct quantitative comparison across all related methods is not always meaningful, the quantitative evaluation here focuses on representative methods with comparable output forms. For the subjective evaluation, 20 participants are invited to rate the naturalness of images generated by the three methods. A set of 10 images is randomly selected, and a five-point scale is used to compute the Mean Opinion Score (MOS) and the standard deviation (SD), which quantify perceptual naturalness and rating consistency, respectively. A systematic comparison among the three methods is performed by integrating subjective results with objective measurements. The overall experimental results are summarized in Table VII.

In terms of objective metrics, the proposed method achieves higher visual naturalness and structural consistency than the traditional occlusion-based method. This indicates that it can

alleviate the visual distortions introduced by occlusion. Compared with the Thumbnail-Preserving Encryption (TPE) method, the proposed method attains higher similarity in the facial regions while maintaining overall image quality. This enhances de-identification and privacy protection effectiveness. In the subjective evaluation, the proposed method also obtains the highest MOS value (4.010) and the lowest SD value (0.742), demonstrating better perceived naturalness and more consistent user ratings. These results indicate that the proposed method achieves superior overall performance in terms of visual naturalness, structural consistency, and perceived quality among representative methods with comparable output forms.

2) *Qualitative Evaluation*: In recent years, adversarial perturbation-based anonymization methods (e.g., Fawkes [41]) have attracted considerable attention in the field of visual privacy protection. These methods embed imperceptible perturbations into the original image to mislead deep face-recognition models during feature extraction, thereby achieving model-targeted identity obfuscation.

Such methods exhibit several notable limitations. First, the injected perturbations typically do not introduce significant visual changes, meaning that the processed images remain easily recognizable to human observers in practical scenarios. Second, their anonymization effectiveness is highly dependent on the architecture and training distribution of the target recognition model, and tends to degrade substantially when confronted with unseen or updated models. Consequently, adversarial perturbation methods are better suited for model-specific defenses rather than privacy-sensitive scenarios that require visually apparent de-identification or model-agnostic robustness.

An additional analysis of face replacement methods is conducted to evaluate the applicability of different categories of privacy-preserving methods. This category includes autoencoder-based face-swapping methods [42] (e.g., FaceSwap) as well as recent GAN-based high-fidelity replacement methods (e.g., SimSwap). These methods share a common characteristic of synthesizing visually natural surrogate faces that preserve the pose and expression of the original images while removing the original identity information. Although such methods can achieve strong anonymization and realistic appearance, they are inherently irreversible. The generated surrogate identities cannot be used to recover the original faces. Therefore, their applicability is limited in scenarios that require data traceability or recovery of the original facial information.

In addition, reversible data hiding-based face privacy methods (e.g., Hiding a Face in a Face [43], RFIS-FPI [44], and F-DDIM [45]) are an important line of comparison because they also support privacy protection with subsequent recovery. These methods are mainly encoding- or embedding-based and usually aim to preserve the visual naturalness of the released image while enabling recovery of the original facial information when required. However, their objective differs from ours. They prioritize reversibility, hidden information security, and minimal visible distortion, rather than explicitly optimizing strong visible de-identification. Consequently, the released image may still preserve considerable facial realism or recognizable facial cues in privacy-sensitive scenarios. In contrast, our method follows

TABLE VII
COMPARISON OF OBJECTIVE AND SUBJECTIVE EVALUATION METRICS ACROSS REPRESENTATIVE VISUAL PRIVACY PROTECTION METHODS

Method	Objective metrics			Subjective metrics	
	Naturalness score	Structural similarity	Face-region similarity	MOS	SD
Obfuscation	0.0059	0.2575	0.2575	1.825	1.172
TPE	0.7579	0.9223	0.8641	1.950	1.169
Our method (block_size=10)	0.8580	0.9463	0.9126	4.010	0.742

TABLE VIII
QUALITATIVE COMPARISON OF REPRESENTATIVE PRIVACY PROTECTION METHODS

Method	Naturalness	De-identification	Recoverability	Resistance to human inspection
Adversarial perturbation methods	✓	×	×	×
Face replacement methods	✓	✓	×	✓
RDH-based face privacy methods	✓	△	✓	△
Our method	✓	✓	✓	✓

Note: ✓ indicates strong capability. △ indicates limited or partial capability. × indicates not supported. Here, de-identification refers to visible de-identification effectiveness, and Resistance to human inspection refers to robustness against human recognition by direct visual inspection.

an AI generation-based paradigm and explicitly targets visible de-identification by suppressing and reconstructing identity-discriminative facial cues. Meanwhile, it preserves natural visual semantics and supports authorized recovery through hidden information.

Overall, these representative privacy-preserving methods differ substantially in terms of visual naturalness, de-identification capability, recoverability, and robustness to human observation. Table VIII summarizes these qualitative differences and highlights the distinct position of our method.

I. Performance Analysis

The runtime performance and hardware resource requirements of the proposed method are evaluated on two hardware platforms, including an RTX 4090 GPU with 24 GB memory and a vGPU server with 32 GB memory. Performance across these platforms is compared to analyze computational latency, memory usage, and practical deployment scope. The average latency of each major processing step, including mosaic generation, super-resolution reconstruction, steganographic encoding, and steganographic decoding, is reported in Table IX, together with the end-to-end latency of the full pipeline.

The results show that the super-resolution stage accounts for the largest portion of the overall processing time, whereas the mosaicing and steganography stages each require only a few tens of milliseconds. Therefore, the main computational cost of the method is dominated by the reconstruction module rather than

TABLE IX
AVERAGE LATENCY OF MAJOR PROCESSING STAGES ON DIFFERENT PLATFORMS

Processing Step	RTX 4090 (ms/image)	vGPU-32GB (ms/image)
Mosaic	45.06	58.43
Super-resolution reconstruction	2766.07	4102.65
Steganography - encode	10.26	13.79
Steganography - decode	27.60	47.17
Steganography - total	37.86	60.96
End-to-end latency of the full pipeline	2848.99	4222.04

by the three-stage pipeline structure itself. In terms of memory usage, the complete process can be executed successfully on both platforms without out-of-memory issues. The current prototype fits within 24 GB of GPU memory. Based on the reported runtime, the method is more suitable for offline or near-real-time privacy-preserving image processing than for latency-critical real-time applications.

VI. CONCLUSION

This paper presents a novel facial privacy protection method that preserves visual usability while ensuring privacy security. It reduces the recognizability of facial images and enhances identity obfuscation, and also enables high-fidelity recovery of the original information. Experiments demonstrate that the method achieves a balanced trade-off among privacy protection, image quality, and reversibility. Compared with existing mainstream methods, the proposed method exhibits superior overall performance in terms of naturalness, structural consistency, and recovery fidelity. These results indicate that the proposed method provides a new solution for facial image privacy protection and serves as a reference for advancing research on privacy-preserving facial image analysis.

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Dezhi An received the BE degree from the Department of Computer Science, Northwest Normal University, Lanzhou, China, in Jun. 1994. He is currently a professor with the School of Cyber Security, Gansu University of Political Science and Law, Lanzhou. His research interests include network security, public opinion analysis, and data mining.



Jingshan Zheng received the BE degree in software engineering from the School of Computer and Software Engineering, Nanyang Institute of Technology, Nanyang, China, in Jun. 2024. She is currently working toward the MS degree with the School of Cyber Security, Gansu University of Political Science and Law, Lanzhou, China. Her research interests include image privacy protection and image steganography.



Ruoyu Zhao received the the MS and PhD degrees in cyberspace security from the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2022 and 2025, respectively. He is currently a lecturer with the School of Computing and Artificial Intelligence, Jiangxi University of Finance and Economics, Nanchang, China. His research interests include multimedia security and privacy. He is also the editor of *Journal of King Saud University Computer and Information Sciences*, and *Expert Systems*.



Yushu Zhang (Senior Member, IEEE) received the PhD degree from the College of Computer Science, Chongqing University, in Dec. 2014. He has undertaken research with the City University of Hong Kong, Southwest University, and the University of Macau. He is currently a professor with the School of Computing and Artificial Intelligence, Jiangxi University of Finance and Economics. His research interests include multimedia security, AI security, and blockchain. He is also an editor of *IEEE Transactions on Dependable and Secure Computing*.



Rushi Lan received the BS degree in information and computing science and the MS degree in applied mathematics from Nanjing University of Information Science and Technology, Nanjing, China, in 2008 and 2011, respectively, and the PhD degree in software engineering from the University of Macau, Macau, China, in 2016. He is currently a professor with the Guangxi Key Laboratory of Image and Graphic Intelligent Processing, Guilin, China. His research interests include image processing and machine learning.